

Lab 1: Linux Programming Laboratory

Code:- IT603 (Introduction to Programming)

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Exercises

1. Create Your Personal Workspace

Create a directory named StudentLab. Inside it, create the following subdirectories:

- Programs
- Assignments
- Notes

Verify the directory structure and display the current working directory.

```
ruchit@Ruchit-PC:~$ mkdir StudentLab
ruchit@Ruchit-PC:~$ cd StudentLab
ruchit@Ruchit-PC:~/StudentLab$ mkdir Programs
ruchit@Ruchit-PC:~/StudentLab$ mkdir Assignment
ruchit@Ruchit-PC:~/StudentLab$ mkdir Notes
ruchit@Ruchit-PC:~/StudentLab$ pwd
/home/ruchit/StudentLab
ruchit@Ruchit-PC:~/StudentLab$ ls
Assignment  Notes  Programs
ruchit@Ruchit-PC:~/StudentLab$
```

2. File Management Practice

Inside the Assignments directory:

- Create the files:
 - assignment1.txt
 - assignment2.txt
 - assignment3.txt
- Do all using Command Lines learned in this Lab.
- Rename assignment3.txt to lab3.txt.
- Delete assignment2.txt.
- Display the remaining files.

```
ruchit@Ruchit-PC:~/StudentLab$ cd Assignment
ruchit@Ruchit-PC:~/StudentLab/Assignment$ touch Assignment1.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ touch Assignment2.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ touch Assignment3.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ ls
Assignment1.txt  Assignment2.txt  Assignment3.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ mv Assignment3.txt lab3.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ rm -r Assignment2.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ ls
Assignment1.txt  lab3.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$
```

3. Creating and Editing Files

Create a file named profile.txt using the Nano editor.

Enter the following information:

- Name
- Enrollment Number
- Department
- Favourite Programming Language

Save the file and display its contents using the terminal.

```
ruchit@Ruchit-PC:~/StudentLab/Assignment$ nano profile.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ cat profile.txt
Name                :- Solanki Ruchit M.
Enrollment Number   :- 202612085
Department           :- MSC.IT
Favourite Programming Language:- C++
ruchit@Ruchit-PC:~/StudentLab/Assignment$ █
```

4. File Copy and Backup

Create a file named report.txt.

Copy it to a new file named report backup.txt.

Edit only the backup file and add one additional line.

Finally, display both files and verify that the original file remains unchanged.

```
ruchit@Ruchit-PC:~/StudentLab/Assignment$ touch report.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ touch report_backup.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ cp report_backup.txt report.
txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ nano report_backup.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ cat report_backup.txt
THIS IS THE LINUX UBUNTU AND WE HAVE TO LEARN
ruchit@Ruchit-PC:~/StudentLab/Assignment$ cat report.txt
ruchit@Ruchit-PC:~/StudentLab/Assignment$ █
```

5. Directory Tree Creation

Create the following directory structure:

LinuxLab

C

C++

Python

Java

Inside each directory, create a file named program1.txt.

Display the contents of every directory.

```
ruchit@Ruchit-PC:~/StudentLab/Assignment$ cd ..
ruchit@Ruchit-PC:~/StudentLab$ cd ..
ruchit@Ruchit-PC:~$ mkdir LinuxLab
ruchit@Ruchit-PC:~$ mkdir C
ruchit@Ruchit-PC:~$ mkdir c++
ruchit@Ruchit-PC:~$ mkdir Python
ruchit@Ruchit-PC:~$ mkdir Java
ruchit@Ruchit-PC:~$ cd LinuxLab
ruchit@Ruchit-PC:~/LinuxLab$ touch program1.txt
ruchit@Ruchit-PC:~/LinuxLab$ cd ..
ruchit@Ruchit-PC:~$ cd C
ruchit@Ruchit-PC:~/C$ touch program1.txt
ruchit@Ruchit-PC:~/C$ cd ..
ruchit@Ruchit-PC:~$ cd c++
ruchit@Ruchit-PC:~/c++$ touch program1.txt
ruchit@Ruchit-PC:~/c++$ cd ..
ruchit@Ruchit-PC:~$ cd python
-bash: cd: python: No such file or directory
ruchit@Ruchit-PC:~$ cd Python
ruchit@Ruchit-PC:~/Python$ touch program1.txt
ruchit@Ruchit-PC:~/Python$ cd ..
ruchit@Ruchit-PC:~$ cd Java
ruchit@Ruchit-PC:~/Java$ touch program1.txt
ruchit@Ruchit-PC:~/Java$
```

6. Directory Management

Create the following directory structure:

Projects

AI

ML

DL

Perform the following operations:

- Delete the ML directory.
- Rename the DL directory to DeepLearning.

- Verify the final directory structure.

```
ruchit@Ruchit-PC:~$ mkdir Projects
ruchit@Ruchit-PC:~$ mkdir AI
ruchit@Ruchit-PC:~$ mkdir ML
ruchit@Ruchit-PC:~$ mkdir DL
ruchit@Ruchit-PC:~$ rmdir ML
ruchit@Ruchit-PC:~$ mv DL DeepLearning
ruchit@Ruchit-PC:~$ ls
AI  DeepLearning  LinuxLab  Python      c++
C   Java          Projects  StudentLab
```

7. Compile and Execute a C++ Program

Write a C++ program that accepts one string and displays is it a palindrome or not.

Compile the program using the GNU C++ compiler and execute it from the terminal.

Record both the compilation command and the program output.

```
ruchit@Ruchit-PC:~/LinuxLab$ nano palindrome.cpp
ruchit@Ruchit-PC:~/LinuxLab$ g++ -o output palindrome.cpp
ruchit@Ruchit-PC:~/LinuxLab$ ./output
enter The Number:- 121

121 is a palindrome
ruchit@Ruchit-PC:~/LinuxLab$ ./output
enter The Number:- 122

122 is not a palindrome
ruchit@Ruchit-PC:~/LinuxLab$
```

8. Git Repository Basics

Create a directory named `GitPractice`.

Inside it:

- Initialize a Git repository.
- Create a file named `readme.md`.
- Stage the file.
- Commit the changes with the message `First Commit`.
- Display the repository status.
- Display the commit history.

```

ruchit@Ruchit-PC:~/LinuxLab$ mkdir GitPractice
ruchit@Ruchit-PC:~/LinuxLab$ cd GitPractice
ruchit@Ruchit-PC:~/LinuxLab/GitPractice$ git init
hint: Using 'master' as the name for the initial branch. This default
branch name
hint: is subject to change. To configure the initial branch name to us
e in all
hint: of your new repositories, which will suppress this warning, call
:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' an
d
hint: 'development'. The just-created branch can be renamed via this c
ommand:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/ruchit/LinuxLab/GitPractice/
.git/
ruchit@Ruchit-PC:~/LinuxLab/GitPractice$ touch readme.md
ruchit@Ruchit-PC:~/LinuxLab/GitPractice$ git add readme.md
ruchit@Ruchit-PC:~/LinuxLab/GitPractice$ git commit -m "First Commit"
[master (root-commit) e402320] First Commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 readme.md
ruchit@Ruchit-PC:~/LinuxLab/GitPractice$ git status
On branch master
nothing to commit, working tree clean
ruchit@Ruchit-PC:~/LinuxLab/GitPractice$ git log
commit e402320869a74714976d97f3bfacef2592dc65e4 (HEAD -> master)
Author: RRUCHIT <onlyruchit06@gmail.com>
Date:   Wed Aug 12 08:05:27 2026 +0000

    First Commit
ruchit@Ruchit-PC:~/LinuxLab/GitPractice$ █

```

9. Mini Linux Project

Create the following directory structure:

Library

Books

Notes

Assignments

Perform the following tasks:

- Create the files:
 - booklist.txt
 - notes.txt
 - assignment.txt
- Copy booklist.txt into the Notes directory.
- Rename the copied file to library books.txt.
- Delete assignment.txt.
- Display the final directory structure.

```
ruchit@Ruchit-PC:~/LinuxLab$ mkdir Library
ruchit@Ruchit-PC:~/LinuxLab$ mkdir Books
ruchit@Ruchit-PC:~/LinuxLab$ mkdir Notes
ruchit@Ruchit-PC:~/LinuxLab$ touch booklist.txt
ruchit@Ruchit-PC:~/LinuxLab$ touch notes.txt
ruchit@Ruchit-PC:~/LinuxLab$ touch assignment.txt
ruchit@Ruchit-PC:~/LinuxLab$ cp booklist.txt notes.txt
ruchit@Ruchit-PC:~/LinuxLab$ mv notes.txt library books.txt
mv: target 'books.txt': No such file or directory
ruchit@Ruchit-PC:~/LinuxLab$ mv library_books.txt notes.txt
mv: cannot stat 'library_books.txt': No such file or directory
ruchit@Ruchit-PC:~/LinuxLab$ rm -r assignment.txt
ruchit@Ruchit-PC:~/LinuxLab$ ls
Books      Library  booklist.txt  output          program1.txt
GitPractice  Notes    notes.txt    palindrome.cpp
```

10. Linux Command Challenge

Create the following final directory structure without using any graphical interface:

Student

Programs

cpp.cpp

python.py

Documents

resume.txt

Backup

resume_backup.txt

```
ruchit@Ruchit-PC:~/LinuxLab$ ls
Books          Notes          notes.txt      resume.txt
CPP.cpp        Programs       output         resume_backup.txt
GitPractice    Python.py      palindrome.cpp student
Library        booklist.txt   program1.txt
ruchit@Ruchit-PC:~/LinuxLab$
```